Responsibilities within the original DT configuration structure

A: specify the decision, quantity, scale, horizon and required evidence.

B: choose how the responsibilities below are implemented. The six families coexist.

Configuration family	Uncertainty responsibility	GeoAI → circular material workflow	Required traceable output
System composition	Retain dependencies when entities are combined	Building / component → stock: shared archetypes and coefficients must remain linked	Joint samples or covariance; aggregation lineage
Internal architecture	Make uncertainty-bearing information traceable	Connect GeoAI attributes, inventory, release, recovery and LCA by stable identities and interface contracts	Entity ID, units, provenance, model version and dependency links
Computational model	Estimate and propagate uncertainty	Infer attributes; compute material mass; forecast release; model recovery; propagate to impacts	Distribution, interval or scenario outputs; stated assumptions
Coupling & control	Test mismatch and route feedback	Compare model output with new survey / inspection evidence; trigger justified downstream revisions	Residual / discrepancy; revision trigger; action permissions
Agency & intelligence	Assign coordination and accountability	Request data, invoke estimators, inspect failed checks and approve recovery decisions	Task owner, escalation rule, decision and audit trail
Temporal organization	Maintain validity over time	Distinguish observation date, model update and forecast horizon; reassess after relevant events	Timestamp, validity interval and affected-output list

C: assign each check to an executable model/service and an accountable human or automated role.

Assess: observe → diagnose → calibrate where justified → validate → propagate → communicate.

A residual flags disagreement; diagnosis establishes plausible causes. Agency coordinates; models compute.

The allocation is our synthesis. Concept references and evidence limits are retained in configuration_responsibilities.csv.